1. Design and develop a responsive web page for your CV using multiple column layouts having video background. You can make the use of bootstrap as well as jQuery.
2. Design and develop a website using toggleable or dynamic tabs or pills with bootstrap and JQuery to show the relevance of SDP, EDI, DT and Course projects in VIT.
3. Develop a website using JavaScript to implement array operations such as reverse the elements of a given array / sorting methods/ searching methods.

4. process user input and manage sessions.

   Task

   1. Design HTML form with Name, Email, Password.

   2. Process form using GET and POST methods.

   3. Validate email format.

   4. Create cookie to store username.

   5. Implement session-based login example.

5. perform database operations using PHP and MySQL.

Tasks:

1. Create database 'student_db'.

2. Create table 'students(id, name, email)'.

3. Connect PHP with MySQL using mysqli/PDO.

4. Insert records into table.

5. Display, update, and delete records.

6. Design and develop a responsive website to calculate Electricity bill using PHP. Condition for first 50 units – Rs. 3.50/unit, for next 100 units – Rs. 4.00/unit, for next 100 units – Rs. 5.20/unit and for units above 250 – Rs. 6.50/unit.

7. Design and develop a responsive website to prepare one semester result of VIT students using REACT, PHP and MySQL. Take any four subjects with MSE Marks (30%) ESE Marks (70%).

   Task

   1. Create multiple components.(App (Parent Component),

   Student (Child Component),Result (Child Component))

   2. Pass data from parent to child using Props.(The App component should pass the data to the Student component are: Student Name,Course,Marks)

3. Manage component State using useState().(Use useState() to manage marks)

4. Update UI dynamically based on state changes.(Display **Pass/Fail status dynamically**)

8. Develop a React-based Student Feedback Form that allows users to submit feedback about a course or session.

Tasks:

1. Create a Feedback Form with Validation

2. Handle controlled components.(Use useState())

3. Use useRef() to access DOM elements.

4. Render list items using Keys.

5. Display Submitted Feedback

9. Develop a **Spring Boot-based Product Inventory Management System** that stores and manages product details using MongoDB as the database.

Tasks:

1. Configure MongoDB connection.

2. Create document class.

3. Create MongoRepository interface.

4.Add Spring Security Dependency

5. Implement Basic Authentication

6. Perform basic CRUD operations.(implement REST APIs for CRUD)

7. Test the Application(Use Postman)

10. Develop a **Student Registration System** where students can submit their details through a Node.js application and the data will be stored in a database. The system should also allow viewing the list of registered students.

Tasks to Perform:

1. Configure the database connection in Node.js.
2. Create a **Student table** with fields like **id**, **name**, **email**, and **course**.
3. Insert new student records using an API or form.
4. Retrieve all student records from the database.
5. Display the student list in the browser using an API endpoint.

11. Design and develop a PHP script to limit the maximum number of concurrent sessions for a user to 3. Set session expiration timeout to 5 minutes.

12. Design and develop attendance systems using PHP and MySQL.

a. student must be able to register himself

b. Teacher should be able to take attendance online using check boxes, roll no and name

13. Design PHP login module with user registration form, login form. System should use cookies to track users. Use session handling and database MySQL for login.

14. Design and develop a responsive website for an online book store using Spring boot and MySQL having
1) Home Page 2) Login Page 3) Catalog Page: 4) Registration Page: (database)

15. Write a web application for registering complaints for students in college. Use PHP and MySQL for frontend and backend.

a) create login page for student

b) create complaint page

c) create login page for admin

d) list all complaints on admin login

16. Develop a currency converter application using ReactJS that allows users to input an amount of dollars and convert it to rupees. Take advantage of React state and event handlers to manage the input and conversion calculations.

17. Write a PHP program to collect waste like plastic or paper. System should accept location where the waste material is present and It should direct concerned authority to collect and manage the waste

18. Write a program in PHP for a complaint management system where users can make complaints about services they are getting from organizations like PMC, PMT or any institution.

19. Write PHP code for booking seats in airplanes and display seating arrangements in airplanes.

20. Write PHP code for Tic-Tac-Toe Game
.
21. Create Responsive web site for showing EDIT and DELETE student records from database using PHP.

22. Create a responsive html document with canvas tag. Write javascript code which takes the mouse click event for the point and draws any shape like line and rectangle.

23. Write html javascript (jQuery) code for applying one style throughout all controls using one of three buttons. Each button is causing separate style for the entire page

24. Write java script code to accept birth date and calculate age in year month and date format

25. Create Responsive web site for showing Traffic signal lights. Use appropriate diagrams and glowing LEDs within HTML code.

26. Develop a **Spring Boot-based Order Management System** that provides REST APIs for managing customer orders. The application should allow users to **create, view, update, and delete orders**.

Tasks:

1.Create Order REST APIs

2.Create Order Entity

3.Test REST APIs(Use **Postman or REST client**)

27. A library wants to store and manage book records in a Node.js application. Librarians should be able to add books and view all books available in the library.

Tasks:

1. Create a **Book table/collection** with fields such as book_id, title, author, and year.
2. Insert book records into the database.
3. Retrieve book data using a Node.js API.
4. Display book details in the browser

28. Design a React application that allows users to **toggle between Light Mode and Dark Mode** using React Hooks.

Tasks to be Performed:

1. Create a toggle button.
2. Use useState() to store the current theme.
3. Change the background and text colors dynamically.
4. Display the current theme mode.
5. Persist theme selection during component re-render.

29. Develop a React application that displays a **real-time digital clock** which updates every second using React Hooks.

Tasks to be Performed:

1. Create a functional React component.
2. Use `useState()` to store the current time.
3. Use `useEffect()` to update the time every second.
4. Display the time in **HH:MM:SS format**.
5. Provide an option to start/stop the clock.

30. Create a React application that allows users to **filter products by category or price range** using Redux to manage filter state.

Tasks to be Performed:

1. Store product data in Redux state.
2. Create actions for **filtering products**.
3. Implement reducer for filter logic.
4. Display filtered products dynamically.
5. Reset filters when required.

31. Build a React application that displays **system notifications** using Redux to manage notification state.

Tasks to be Performed:

1. Create Redux store for notifications.
2. Implement actions for **Add Notification and Remove Notification**.
3. Create reducer to update notification state.
4. Display notifications in the UI.
5. Allow users to dismiss notifications

32. Develop a Spring Boot application where user passwords are **securely stored using password encryption**.

Tasks to be Performed:

1. Configure password encoder (BCrypt).
2. Store encrypted passwords in database.
3. Authenticate users with encrypted passwords.
4. Verify password validation during login.
5. Display authentication results.

33. Create a Spring Boot application that tracks and restricts **multiple failed login attempts** to enhance application security.

Tasks to be Performed:

1. Configure Spring Security authentication.
2. Track failed login attempts.
3. Lock account after several failed attempts.
4. Display appropriate error messages.
5. Allow account unlock after a specific time.

34. Create a **blog management REST API** using Express.js where users can create, read, update, and delete blog posts.

Tasks to be Performed:

1. Setup Express server.
2. Create routes for **creating, viewing, editing, and deleting blog posts**.
3. Store blog data in memory or database.

4. Return responses in JSON format.
5. Test the APIs using Postman.

35. Build a **Task Manager REST API** using Express.js to manage daily tasks.

Tasks to be Performed:

1. Create API routes for **adding tasks**.
2. Retrieve all tasks using GET requests.
3. Update task status (completed or pending).
4. Delete tasks when completed.
5. Return task data in JSON format